

Vanessa Ching

U.S. Citizen · vanessazqching@gmail.com · (678) 779-0547 · [linkedin.com/in/vanessa-ching](https://www.linkedin.com/in/vanessa-ching)

EDUCATION

Johns Hopkins University, Whiting School of Engineering

Baltimore, MD | Aug 2024 – May 2028

B.S. in Computer Science, Minor in Film and Media Studies | GPA: 3.28

Relevant Coursework: Computer Systems Fundamentals, Intermediate Programming, Gateway Computing: Python, Data Structures

EXPERIENCE

Georgia Tech Research Institute (GTRI) · Research Intern

Atlanta, GA | May 2026 – Aug 2026

- Integrating Unreal Engine and computer vision pipelines (OpenCV, YOLO) into hyperrealistic synthetic simulation environments with KLV/UAS metadata support for autonomous perception testing
- Developing a multi-environment vehicle tracking system evaluating AI perception across photorealistic terrain, lighting, and occlusion conditions

Sports Analytics Research Group, JHU · Undergraduate Researcher

Baltimore, MD | Jan 2026 – Present

- Built Voice Scorekeeper, a real-time AI baseball scorekeeping platform using React, TypeScript, Node.js, and Tailwind CSS
- Engineered a low-latency backend pipeline using OpenAI-compatible LLMs (OpenAI, Ollama, Groq) to convert spoken play-by-play commentary into structured game events
- Developed interactive spray chart and live game-state visualizations with human-in-the-loop confirmation workflows for scoring accuracy
- Optimized inference and transcription pipeline latency from ~21s to ~7s through prompt reduction, response optimization, and lightweight local model support

Social Cognitive AI Lab, JHU · Research Assistant

Baltimore, MD | Aug 2025 – Jan 2026

- Developed procedural urban simulation environments in Unreal Engine 5 and Python for embodied AI navigation and perception experiments
- Designed dynamic multimodal scenarios incorporating environmental variation, moving obstacles, and spatial audio cues to support agent reasoning and navigation research

Frenalytics · Software Engineering Intern

Remote | May – Aug 2025

- Built an AI-powered WordPress reading assistant platform supporting 1,500+ students with literacy challenges
- Designed PHP REST APIs and MySQL-backed session infrastructure for avatar customization, progress tracking, and educator analytics
- Implemented scalable backend features emphasizing maintainability, structured data storage, and WordPress ecosystem compatibility

PROJECTS

Voice Scorekeeper | React, TypeScript, Node.js, Express, Tailwind CSS, LLM APIs

- Real-time baseball scoring platform combining speech transcription, LLM-based play parsing, live game-state tracking, and spray chart visualization
- Supports OpenAI-compatible APIs and local Ollama inference for low-latency offline workflows

AI Readezzy Reading Assistant | PHP, MySQL, JavaScript, WordPress

- AI-assisted reading platform featuring dynamic avatar systems, session persistence, REST APIs, and relational database-backed personalization workflows

LEADERSHIP & COMMUNITY INVOLVEMENT

Pi Beta Phi Fraternity for Women · Vice President of Operations

Baltimore, MD | Sep 2025 – Present

- Coordinate chapter operations and cross-functional officer initiatives for a 60+ member organization

Hopkins Student Organization for Programming (HOP) · Fine Arts Co-Chair

Baltimore, MD | Sep 2024 – Present

- Co-lead logistics and vendor coordination for campus-wide fine arts programming serving 5,600+ students

Matriculate · Advising Fellow

Remote | Jan 2026 – Present

- Mentor high-achieving, low-income high school students through the college application process

Student Government Association · Class Council Representative

Baltimore, MD | Jan 2026 – Present

- Execute large-scale class initiatives for a cohort of 1,000+ students; Sophomore Rep (2024–25), Junior Rep (2025–26)

SKILLS

Languages: Python, C++, C, Java, JavaScript/TypeScript, SQL, PHP

Frameworks & Tools: React, Node.js, Express, Tailwind CSS, MySQL, REST APIs, Git, Linux, WordPress

AI & Simulation: OpenCV, YOLO, Unreal Engine 5, Unity, LLM APIs (OpenAI/Ollama/Groq)